

Queue Management in College Canteens: A Participatory and Observational Approach to Understanding Waiting Experience and Process Delays

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Abstract

Queuing is a central and insufficiently studied aspect of service operations, particularly where the demand periods are short with high intensity. College canteens can be considered such environments in which students and faculty members face strong time limitations in limited academic breaks. The traditional queue management approach mostly focuses on minimizing the waiting time, often ignoring experience-based factors that will breed dissatisfaction even when under minor delay conditions.

The current case study examines the practice of managing queues in five canteens of colleges based on a participatory, observational research design. The systematic observation was conducted on two hundred and fifty structured observations at the peak and non-peak sessions and the customer behaviour, affective response, and process flow were recorded rather than relying on self-reported information. The level of satisfaction was estimated by means of a Likert-type observational scale based on behavioural indicators that could be observed.

The results show that dissatisfactions cannot be attributed only to the real period of wait times, but rather to other aspects like congestions, uncertainty, idle standing, lack of information and loss of control. The demand peaks and the resulting frustrations were intensified by the focus in busy hours, and the introduction of digital display of the orders and more transparent process designs significantly reduced the emotional flare-up. In a conclusive manner, the research confirms that queue management in college canteens is fundamentally a human factor service-design problem, which places more emphasis on predictability, information visibility and sense of experience over increased capacity.

Keywords:

Waiting experience, queue management, participatory observation, service design, college canteens.

1. Introduction: Queue Management and Its Importance.

Operation of queues is essentially a part of service operations in a system where the demand of service surpasses the capacity of service delivery upon demand. In industries like hospitals, banks, retail outlets, transport systems, restaurants and schools it is inevitable to find queues. Nevertheless, the design and communication of queues and how they are handled is a decisive factor in determining the level of user satisfaction, operational efficiency and service perception, in general.

Badly managed lines do not only create an increased waiting time. They cause mental strain, exasperation, sense of injustice, and adverse affective reactions among users. In the long run, the continued negative waiting experiences will be able to undermine trust in service system and deter usage of the service system, despite the fact that the core service output may technically suffice.

College canteens can be considered a very complicated queue environment. Students and faculty are accessible in huge numbers during short academic break periods, and are usually on a rigid time schedule of getting back to classes or conferences. These states impose high demand that is brief but extreme. In this case, even minor inefficiencies, information deficits, or even slight delays will have an imbalanced impact on customer experience.

The conventional service design methods tend to believe that waiting time alone will be reduced, which will enhance satisfaction. But experience shows that people react not only to the duration of waiting, but also the structure of the waiting, the predictability of it, the congestion of the overall space and the amount of information or control that they perceive during waiting. This case study aims at these experiential and behavioural aspects of waiting that most traditional queue analysis tries to ignore.

2. Research objectives

The given study was carried out in accordance with the well-formulated goals that regulated data gathering, analysis, and interpretation. These goals will guarantee that the study goes beyond mere observation and that it leads to practical knowledge.

The specific objectives are:

- To explore the difference in the experience of queuing during peak and non-peak time in college canteens with special focus on behavioural and emotional reactions.
- To determine how the non-time characteristics of uncertainty, congestion, visibility, social interaction and urgency predict dissatisfaction in waiting.
- To determine how the design aspects of the process such as visibility of the kitchen, electronic displays of orders and self-order systems influence customer behaviour and perception of waiting.
- To learn the difference in the experience of various classes of users (individuals, groups, students, faculty) of the same queue system.
- To create flexible, empirically sound knowledge on how to enhance the management of queues with service design and continuous feedback instead of increasing capacity on its own.

3. Conventional Methodologies of Waiting Time Measurement.

Majority of the current queue management research is based on mathematical modelling, analytical modelling or simulation. The models consider quantifiable variables like arrival rates, service rate, average waiting time, queue length, throughput and system utilization. These methods are useful in capacity planning, staffing and prediction of congestion.

These models however assume a relatively constant operating environment and homogeneous customer behaviour that is rational. These assumptions are unsuccessful in college canteens. There is a high level of demand volatility, with sharp fluctuations in demand within a small period of time, the speed of service delivery is determined by the type of order, and urgency, emotion, social context, and environmental stress determine customer behaviour.

Consequently, two queues having similar average waiting time might seem entirely different to customers based on the size of the crowd, visibility of orders, or even seem fairly. Traditional models describe the waiting duration of people, yet they fail to explain why individuals get frustrated, anxious, or relaxed throughout the time of waiting. This drawback underscores the significance of methods that are able to directly represent human experience.

4. The case of why Participatory and Observational Research is necessary.

Participatory and observational research presupposes direct interaction with the environment of the service and close observation of actions according to the course of their implementation in a real scenario. This method does not rely on the retrospective surveys or abstract averages, but rather records visible reactions, interaction and escalation of emotions in real-time.

Dissatisfaction in a queue setting is usually manifested non-verbally, which can be pacing, checking, asking personnel, evident annoyance, or disinterest. The observational methods thus can be well applied in the study of the behaviour of queues.

The researcher can obtain an insight into such subtle yet important dynamics as congestion spillover, fixation of attention, and emotional escalation, which cannot be observed with the help of numerical data only.

5. Problem Statement

College canteens are often faced with a lot of dissatisfaction during peak times despite having sufficient food preparation and trained personnel. Some of the common problems experienced are:

- Long and congested queues
- Obvious impatience and frustration.
- Constant inquiring and grumbling.
- Delay in order fulfilment and some errors.
- Uneasiness on the part of faculty and top management.

The main issue is not the period of waiting, but the design, communication of the waiting process, and the experience of it with severe time pressure.

Research Question:

What is the rationale behind the fact that waiting lines in college canteens can create a significant amount of dissatisfaction even under the condition that differences in the actual service times are relatively minor?

6. Context of the Cases and Study Environment.

The research was carried out among five college canteens that had different operational designs:

Canteen 1: Open kitchen, no computer display, no self-order system.

Canteen 2: Kitchen Open, no digital display.

Canteen 3: Semi-open kitchen with no digital display.

Canteen 4: Digital display order with kitchen closed.

Canteen 5: Self-order display kitchen towards the counter.

The observations were made in peak and non-peak times (lunch and snack breaks). A total of 250 structured observations were carried out using an observation sheet that was predetermined using excel. This was an adequate sample size to determine common behavioural patterns across time and canteens.

7. Procedure: Data Analysis and Observational Design.

The research methodology used in this study was a non-invasive method based on observational research where a participatory approach was employed to study the behaviour of queues and waiting experience in college canteens in real working conditions. The reason behind employing the research methodology adopted was informed by the nature of the research problem which focuses on the understanding of human experience, behavioural responses and psychological responses during waiting, as opposed to measuring operational performance measures.

Explicit complaints are not common in queue environments in which dissatisfaction is expressed. Rather, it presents itself in form of tiny behavioural indicators like body language, checking the service counter incessantly, pacing, audible sighs, evident frustration, disinterest or talking to the personnel repeatedly. These expressions tend to be informal and not foreseeable hence may not be easily captured by using surveys or interviews that are carried out after the experience is over. Observational studies therefore suit best in the context of observing real-time emotional and behavioural dynamic.

It was studied in natural environments without the manipulation of staffing, physical layout, service processes and technology use. This was done to make sure that the data contained genuine queue behaviour and not responses to experimental manipulation. The researcher was passive during the observation period so that he appeared to be a part of the environment to reduce the impact of the observer.

7.1 Type of Study

This research adheres to a qualitative-dominant observational research design, which is backed by structured quantitative summaries that were based on the behavioural scoring. It is both descriptive and analytical instead of being experimental or predictive.

The customers were not interviewed, surveyed, or even told that they were being observed. The reason behind this approach was to prevent bias in responses and social desirability effects which may skew self-report satisfaction statistics. Rather the study is based on what people do and not what they claim to feel.

The study did not aim at testing a hypothesis in a statistically significant manner, but to:

- Determine the patterns of behaviour in queue conditions.
- Knowledge of why discontent intensifies in special circumstances.
- Connection between behaviour and queue design and service context was noted by Link.

This would be consistent with the exploratory service research in which the interpretation of lived experience is done before the formal testing of models.

7.2 Observation Framework

To achieve consistency of data between canteens and the time, a structured observation framework was already created before the data were collected. The several dimensions of the queue experience measured in each observation were:

- The development and movement of queues.
- Physical overcrowding and interpersonal distance between customers.
- Signs of calmness, impatience, irritation, aggravation.
- Frequency and type of contact with service employees.
- Adoption of a supporting system like self-order kiosk or digital order display.
- The contextual conditions of when it is peak or non-peak and time pressure in view.

They were observed in five college canteens of differing designs of operation during the high-peak hours (lunch and snack breaks) and low-peak hours. This allowed comparative examination under different conditions of demand and queue arrangement systematically.

A total of 250 structured observations were filled in an existing predefined Excel based observation sheet. This number of observations was enough to discover common patterns of behaviour in different locations and time which is suitable to qualitative-dominant observational studies.

7.3 Analysis Technique

The analysis of data was done in a pattern-matching and thematic comparative analysis method.

First, the observations were grouped on major analytical dimensions:

- Peak versus non-peak periods
- Canteen design type
- Availability or unavailability of digital order displays.

- Contexts of individual waiting versus group waiting.

Second, behaviour patterns were observed in each of the groups. The same behaviours invariably observed under certain circumstances, like pacing more in busy periods, or acting calmer in canteens with displays on them, were considered context driven phenomena and not isolated events.

Third, the behavioural Likert scale was used to produce descriptive quantitative summaries including average satisfaction scores, frequency distributions. Inferential statistical testing was not done using these summaries, but to justify and visually confirm qualitative interpretations.

This combination of interpretive methods was necessary to make sure that both observed behaviour and structured evidence were the basis of the findings, which enhanced the credibility of the analysis.

Microsoft Excel was used to create data organization, descriptive summaries, and visual representations. Observation records were organized with the help of Excel, the average scores of the satisfactions were calculated, and the comparison charts were created to contribute to the interpretation of the observations.

7.4 Satisfaction as an operational definition

The behavioural scale was developed as a Likert-based scale of 1 to 7 that is based on visible behavioural information, and it operationalized customer satisfaction.

The lower scores were relaxed and disengaged behaviour whereas the higher scores were increasing dissatisfaction which was full of restlessness, checking repeatedly, verbal complaints, or emotional outbursts. These scores were not to be used and analyse individual customers but to provide the system level analysis.

Analytically, a derived churn-risk indicator was prioritized to identify areas of improvement and it was determined that negative waiting experience repeated multiple times may impact on future use in semi-captive situations like college campus.

7.5 Observational Data and Variables Description.

The observational data of this case study were recorded with a structured observation fieldwork based on an Excel based observation sheet that was developed before going to the field. The data in the dataset was in the form of a row which was an instance of a queue observation at one of the five college canteens, whether it was a peak period or non-peak.

The key variables in the observation sheet were as follows:

- Canteen identifier (Canteen 1 to Canteen 5)
- Time category (Peak / Non-Peak)
- Level of congestion at the queue (Low / Medium / High)
- Availability of digital order display (Yes / No)
- Open / Semi-open / Closed Type of Kitchen visibility.
- Waiting (Standing / Mixed / Seated) posture.
- Waiting situation at individual or group level.
- Behavioural changes (restlessness, pacing, complaints, calm behaviour) are observable.

- Inferred satisfaction score (Likert based).

The data were 250 structured observations, which were allocated in canteens and periods to allow enough exposure to repeated behavioural patterns. The data were not to be subjected to inferential statistical testing but were to be utilized in the support of the descriptive summaries, visual comparisons, and identification of patterns.

This systematic data base allowed comparative analysis of canteens and demand conditions systematically and included the qualitative richness of the participative observational research.

8. Queue Experience Conceptual Model.

Using recurring participatory studies on five college canteens, this study hypothesizes a conceptual model of queue experience which indicates dissatisfaction as a result not of an explicit relationship between waiting time directly, but rather of interacting design, contextual, and psychological influences.

The classical queue models consider waiting as a linear waiting between arrival and service delivery. But this study has observed that waiting is not perceived by the customers to be one continuous unit of time. Rather, waiting is perceived as a series of steps, each of which has its emotional reaction and perceived control.

The conceptual model created during this research conceptualizes the concept of queue experience as a service journey through multi-stages, which includes:

- Order placement
- Payment
- Waiting for preparation
- Order pickup
- Exit

The various stages present various psychological states. These phases are determined by the design of queues and its context of service i.e. layout visibility, visibility of their kitchen, use of technology, and staff conduct. These design factors are further moderated by situational factors, such as the intensity of peak demand, the urgency with respect to the academic schedules and social context (waiting individually or in groups).

A combination of these factors influences the waiting experience, and it is defined by:

- Levels of certainty or uncertainty.
- Congestion and crowd level.
- Extent of idle standing
- Sense of control of the process.

The end product of such an experience is a feeling of observable contentment reactions, which could be a state of placid acquiescence to open frustration.

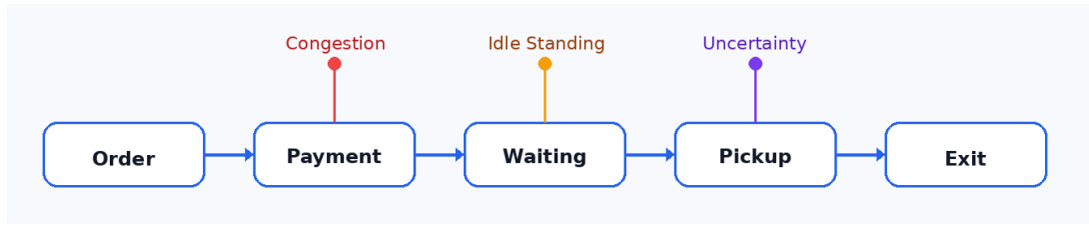


Figure 1: Process and Stress Point of Queue and Experience.

The figure 1 presents the service flow (Order → Payment → Waiting → Pickup → Exit) visually and indicates the essential points of stress, such as congestion, idle standing, and uncertainty as a result of observation. This number is used to provide the analytical basis on which the empirical results provided in subsequent sections are interpreted.

9. Findings of the Study

This part provides the main empirical results based on the 250 structured participatory observations, which were carried out in five college canteens. The results are structured in themes to derive how dissatisfaction develops, gets worse and diffuses in varied circumstances. This is not simply about determining dissatisfaction, but how it works.

9.1 Peak-Hour Demand Concentration as the Main Breadwinner of Dissatisfaction.

In all five canteens, there was a general trend observed, which was an increase in dissatisfaction during the peak hours regardless of the continuity of service flow. High customer congestion, noise, lack of sufficient personal space and high time pressure associated with academic work made it peak hours.

It was observed that customers paced frequently, repeatedly stared at the counter, verbal sighs, and displayed impatience even in the moving queue during peak hours. The overwhelming reactions were caused by short pauses in operations e.g. when staff were changing activity or when bulk orders were involved.

Equally, the same operational pause during off-peak hours hardly attracted any obvious annoyance. Even when waiting time was equal, customers seemed to be relaxed, interacted with their phones, or had a conversation in a calm manner.

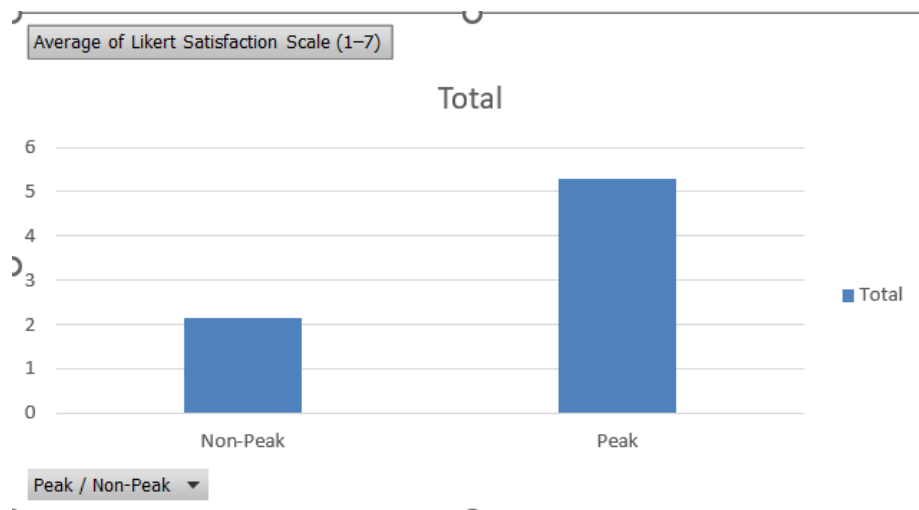


Figure 2: Likert Satisfaction Mean Score- Peak vs Non-Peak.

Here in Figure 2 the number is a quantitative measure of the dramatic difference between peak and non-peak times, which supports the conclusion that the concentration of demand and surroundings is more likely to induce dissatisfaction than absolute waiting time.

9.2 Canteen-Wise Differences and Process Design Role.

Despite the dissatisfaction during the peak hours in all the canteens, it was identified to be very strong in terms of its degree of variation based on the clarity of the processes design. Canteens that had well defined ordering, waiting and picking up areas had less emotional escalation than those with a non-clear layout.

In canteens where the customers could be visually and spatially aware of their standing points, waiting period, and receiving order position, the queues and customer-related behaviour were less chaotic. On the other hand, ambiguous process boundaries resulted in the customers clustering around counters entailing congestion loops which intensified stress.

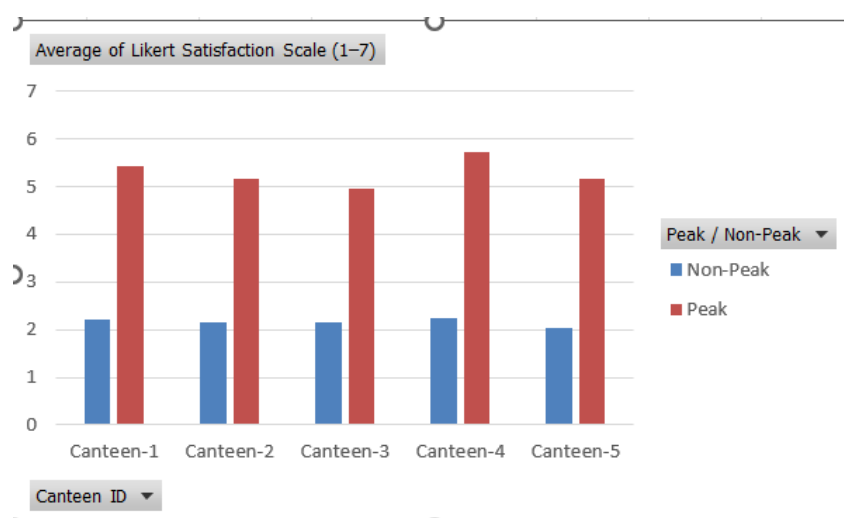


Figure 3: Canteen-Wise Comparison of the Satisfaction Scores.

In figure 3 the value illustrates that design disparity is reflected in quantifiable disparities in perceived satisfaction, despite the same level of demand pressure.

9.3 Digital Displaying the Orders and Psychological Control.

One of the most useful moderators of waiting stress proved to be digital order displays. Customers showed increased trust in the system in canteens with displays, moved away counters, and were waiting in a passive manner.

Order numbers offered psychological control to customers, although the waiting time could be the same. The screen helped to remove uncertainty since it responded to the implicit question: “Is my order moving?”

Canteens with no displays, customers kept asking questions, over and over again to the staff, watching kitchen processes, and being physically in the vicinity of the counter, congesting and straining emotionally.

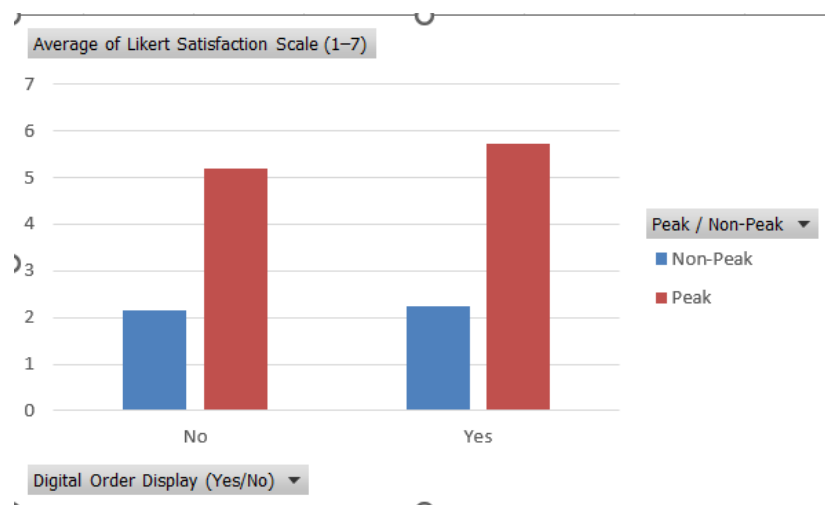


Figure 4: Effects of Digital Order Displays on Satisfaction.

This value demonstrates the stabilizing nature of the information visibility on waiting behaviour.

9.4 Extravagant Frustration and Affective Contagion.

Dissatisfaction scores (6–7) were almost focussed during peak hours (high scores). Notably, such radical reactions tend to permeate outside the person who is initially affected.

In the cases where a customer broke out in frustration, e.g, by screaming, losing his temper with employees, customers stood closer to him and were more alert, tensed and impatient. This effect of emotional contagion only contributed to the general queue atmosphere, where small problems seemed structural.

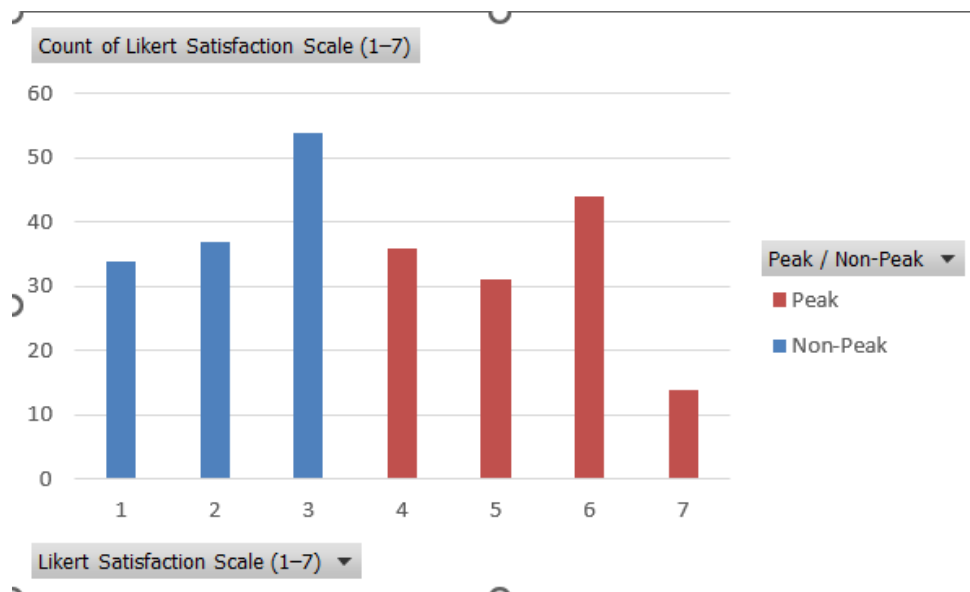


Figure 5: Likert Satisfaction Scores distribution by Period.

This figure 5 demonstrates that extreme frustration is not random but a situation, which supports the necessity to avoid high-impact incidences.

9.5 Individual/Group Waiting Dynamics.

The difference was found between single and group waiting customers. Those who were not waiting with others were found to be engaged in observing the queue in a continuous manner paying close attention to the counter action and order status. The result of this sustained attention was faster emotional escalation, and visible indicators of emotional escalation, including restlessness, pacing, and repeated checking.

Customers on queues on the contrary looked more relaxed. Interaction in groups caused the loss of attention to the queue, which led to a perception of lower intensity of waiting, despite similarity in actual waiting time. This shows that perceived delay is lower in the case of social interaction and dissatisfaction is evened out.

9.6 Effect of External Time Pressure and Academic Urgency.

Customers who had immediate academic engagements showed more anxiety when they were waiting. These people kept on looking at the clock and going to employees to make inquiries and were visibly impatient even when the service was flowing fairly. The feeling of dissatisfaction was experienced at an early stage of waiting process and in most cases, earlier than delays became a major concern.

This conduct is in contrast to the customers who did not have a pressing time schedule and stayed calm under such circumstances. The result indicates that urgency increases dissatisfaction regardless of the real waiting time.

9.7 Error of Orders and Reliability Sensitivity during peak hours.

Order mistakes were very rare but were very powerful during peak time. Wrong orders or a slow response resulted in visible aggravations and multiple contacts between the staff, which could attract the attention of other customers. These accidents heightened tension in the total queue.

The same mistakes were treated at non-peak periods with relative calmness which implied more tolerance. This observation emphasizes the fact that reliability of service is more important than speed in creating customer satisfaction during peak demand.

10. Discussion: Findings Integration by a Lens of Behavioural Service Design.

The overall results of the study point to the fact that the level of dissatisfaction with the queues in the college canteens is not a linear effect of the waiting time, but rather, a resultant behaviour due to the interaction of psychological stressors. Congestion, uncertainty, having nothing to do, social setting, urgency and reliability failures do not act on their own. Rather, they inter-relate to produce emotional stimulation in peak demand periods.

One insight that is critical is the perceived control. Customers who were made to feel that they are informed, social and confident about the progress of their orders were not as anxious even in long waiting lines. On the other hand, the customers who appeared to be uncertain, lonely, or in a hurry, became frustrated quickly. This is the reason why technological interventions like digital order displays worked- not because it helped to reduce waiting time but because there was a sense of predictability that was filled.

It has also been shown in the discussion that queue dissatisfaction is contagious. Emotional outburst by one person tended to change the emotional mood of the other end of the queue. This multiplier impact implies that a handful of high-frustration incidences can have a disproportionately large impact on the customer experience on the whole.

These findings, as seen through the prism of service design, imply that the perspective of an efficiency-driven model is reduced in favor of the experience-oriented queue design, in which emotional control is being approached as an operating goal.

11. Implications of management towards Canteen operations.

As a canteen manager, the research has a number of practical implications. First, information visibility (such as digital displays or order-status boards) investments are high in terms of experiential returns with a relatively low cost. Second, physical redesign which distinctly separates the ordering, waiting and pickup area lessens frustration due to congestion.

Third, the managers are to perceive peak hours as a different mode of operation and not an expansion of the regular service. Peak-hour menu, staffing roles and error prevention mechanisms are available to be simplified and standardized to reduce the reliability failures.

Lastly, it is necessary to incorporate a continuous observation and short feedback loops into everyday functioning, which would enable the managers to spot developing stress points in time before it becomes overwhelming.

11.1 Recommendations on Proactive Based on Observations.

According to the results of this case study, it is possible to suggest the following practical recommendations on how to enhance the experience with queues in college canteens:

- Enforce physical distinction between ordering, waiting and pickup areas in order to minimize congestion and cognitive load.
- Install electronic order display to minimize confusion and unnecessary staff disturbances in periods of high demand.
- Discourage waiting in a single position, as far as possible, to prevent excessive standing and physical exhaustion.
- Support peak hours as a different operating mode having simplified menus and standard operating procedures to minimize order errors.
- Schedule the position staff jobs during high-demand times to explicitly ensure precision of the orders and customer contact instead of just speed.
- Implement optional and fast feedback systems at the exits to facilitate the continuous monitoring of the experience.

The recommended ones focus more on service design and behavioural control as opposed to capacity expansion and thus are practical in an institutional context with limited resources.

11.2 Implications at an Institutional/Policy Level.

On the institutional level, the results present that the problem of queuing congestion is not a canteen issue only, but a problem related to the system coordination. The releases of student masses at a time in academic schedules generate unnatural demand bursts affecting service systems.

The mitigation measures that can be employed by institutions would include staggering of break times, extending the break period by a few minutes during the peak semesters, or pre-order systems that are combined with academic schedule. There is also some faculty discomfort in shared queues, which indicates that differentiated service arrangements are necessary.

The consideration of queue experience as campus service quality prompts a more comprehensive student and faculty satisfaction.

12. Manipulations and Limitations of the Study.

This paper is specifically dedicated to queue experience and process flow in the college canteens. It does not look at pricing policies, menu sophistication, employee wage expenses, nutritional value, or contract-related factors with the vendors.

Methodologically, satisfaction was based on observable behaviour and not on the self-reported perceptions. On the one hand, this is reflective of real time reactions, but it is not necessarily a complete picture of internal attitudes or long term behavioural interventions. Also, time-stamped service statistics are not available, so it is impossible to directly compare the performance of the traditional queue with them.

12.1 Assumptions of the Study

The research was carried out on the following assumptions which are recognized to explain boundaries of interpretation:

- The customer behaviour under observation is an indicator of perceived waiting dissatisfaction although internal perceptions are never given a verbal expression.
- Patterns of peak-hour demand are comparable enough in the case of canteens to permit contextual comparison.
- The observer did not make a big difference on natural queue behaviour because of non-intrusive and passive method of observation.
- There were no coordinated actions among observation periods and customers waiting in queues without coordination.
- Environmental and operational conditions During the period of observation, the conditions were typical of normal academic days.

These assumptions are in line with the known observational research practices and were needed to make behavioural patterns meaningful in real-life service settings.

This study is described as a situational case study, and it is not pushed towards statistical generalization. The results are expected to offer analytical knowledge and transferrable learning instead of general predictability. Different interpretations must therefore be put into use with local considerations and demand trends and institutional limitations.

13. Validity, Reliability, and Methodological Rigor.

Validity and reliability are also essential issues that need to be ensured in observational and qualitative-dominant research. Observational studies are credible because, unlike purely quantitative studies, where rigor may be established by statistical tests, observational studies have to be conducted with systematic design, consistency, and transparency. The present work took care of these issues with the help of various methodological precautions.

13.1 Internal Validity

The internal validity is the ability of the findings to reflect the phenomenon of study. The internal validity in this study was enhanced by the pattern repetition in contexts. The same behavioural reactions, including, behavioural impatience in the busiest of the days, behavioural calmness with the digital displays, emotional outbursts as a result of order errors, were observed to be similar across the various canteens and days. The repetitive patterns decrease the chances of incidental and context-dependent findings.

Also, arbitrary interpretation was reduced due to the utilization of a pre-defined observation framework. The indicators that are behavioural in nature pacing, checking repeatedly, verbal complaints, and disengagement were all determined ahead of time and the observations were based on uniform criteria as opposed to judgments that are post-hoc.

13.2 Reliability

Reliability has to do with the consistency of measurement. In our research, reliability was achieved by using a structured observation sheet in the form of Excel sheets that were applied similarly to all the 250 observations. Observation categories and recording method as well as identical behavioural scale were applied regardless of canteen type or time of the year.

Although the replicability of qualitative observation cannot be attained to its fullest extent, the clarity of observation framework assures the other researcher working within the same context and under the same criteria of observation that a similar pattern would predominantly be detected.

13.3 Construct Validity

Construct validity concerns the aspect whether the study is measuring what it purports to measure. The measure of satisfaction was not based on self-reported opinion but derived based on observable behaviour. Although this method does not measure internal cognitive appraisals, it effectively reflects felt dissatisfaction as it occurs in real time which is precisely what the study aims to accomplish in its objective of getting a better understanding of the waiting experience as opposed to reported attitudes.

13.4 Limitations to Rigor

There is no triangulation between perceived experience and objective metrics due to the lack of surveys or time-stamped service data to align them. This weakness is accepted and discussed in the future research section.

14. Ethical Considerations

Ethical responsibility was also an issue that was given serious consideration during the research especially since the research was an observational one. The observations were done only in open service locations where one can observe the queue behaviour at all times, there is no breach of expectation of privacy.

Any personal data were not gathered during any phase of the study. This will contain names, photos, video recordings, facial information or any other characteristic of customers or employees that identifies them. The analysis was entirely based on anonymous, aggregated behavioural observation, without any effort to monitor individuals over time and encounters.

No interaction or observation was done between the researcher and the participants, no information was given to the customers that they were observed. This non-interfering method was necessary in order to not interfere with the behaviour and keep the waiting experience in its original form. Notably, there was no deception; this means that the study did not control or mislead the participants.

Only analytical tools, i.e. satisfaction scores and churn-risk indicators, were employed to determine stress points at the system level. They did not apply to label, assess or judge either individual customers or staff performance.

The paper also highlights that future application of digital queue systems, analytics and feedback applications should be based on the principles of:

- Voluntary participation
- Transparency of purpose
- Minimal data collection
- Respect for user autonomy

The use of ethical queue management ought to improve experience without ushering in surveillance and coercion.

15. Contributions of the Study

This paper is valuable at three levels, including theoretical, methodological, and practical levels.

15.1 Theoretical Contribution

Theoretically, the research is an expansion of classical theory in queue management as it shows that the waiting experience is not a time-based phenomenon but a behavioural and psychological phenomenon. It questions the superiority of time-related measurements of efficiency by demonstrating that uncertainty, congestion, perceived loss of control rather than waiting time itself lead to dissatisfaction.

The humanistic model that is formed during this research offers a humanistic approach of paradigm to comprehend queues, especially in high pressure, urgent institutional settings.

15.2 Methodological Contribution

The study methodologically proves the usefulness of participatory observational research, which is integrated with structured behavioural scoring. The strategy lies between qualitative observation and quantitative abstraction and is a pragmatic way to study real-world service systems and not interfere with their functioning.

The application of the Excel-based structured observation also indicates how ready-made tools could be used to conduct systematic research on the service.

15.3 Practical Contribution

In practice, the research provides relatively inexpensive, design-driven interventions which institutions need not invest significant resources to adopt. Some of them are to enhance information visibility, service stage separation, and seated waiting, and reliability at peak hours.

The results can be directly applied to the canteen managers, campus administrators, and service designers.

16. Future Research Directions.

Although this research offers a superb insight into the nature of queue experience at college canteens, there are still a number of opportunities that can be explored in research.

Further research may involve addition of time-stamped service data in order to establish a direct comparison between perceived waiting experience and actual waiting time. This would enable a greater integration of behavioural insights and the traditional queue models.

The generalizability of the research would be enhanced by conducting the study in more campuses, institutions, or cultural settings, which would demonstrate the effects of institutional norms on waiting behaviour. Redesigned queue systems, e.g., seated waiting zones or pre-order mechanisms, could also be tested in experimental studies and the effect of these systems on satisfaction weighed.

The longitudinal study might investigate the effect of repeated waiting experiences on long-term behaviour, e.g. avoidance, timing changes, alteration of service usage pattern.

17. Conclusion and General Reflection.

As it is shown in this case study, the management of queues in college canteens is not an issue of the operational efficiency, but rather a human-centered service design problem. Dissatisfaction is not necessarily about waiting time, but the experience of waiting, when there are congestion, uncertainty, urgency, and social context conditions.

Through the participatory and observational research strategy, the study exposes aspects of waiting that are not visible in the traditional queue models. The results indicate that customer experience during peak demand can be greatly enhanced through comparatively simple interventions, which are oriented at information, predictability, and comfort.

Finally, the proper running of the queue demands constant monitoring, dynamic design, and ethical responsiveness. Service systems that are based on the actual behaviour of human beings instead of abstract assumptions are more resilient, equitable, and sustainable.